$S(6) \le 1801$ if $R_4(3) \le 61$: a centred Schur bound and the structure at the frontier

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Abstract

Let $S(n)$ be the Schur number and $R_k(3)$ the Ramsey number of the triangle in k colours. We prove that $R_k(3) \le r$ implies $S(k+1) \le 2(k+1)\lfloor (r-1)/2 \rfloor + 1$. For even r this is k less than the bound $(k+1)(r-1)$ that the pigeonhole recursion and the inequality $S(k+1) \le R_{k+1}(3) - 2$ give. Section 5 discusses the relation of this bound to a Schur bound of H. Wan (1997), whose proof we have not read. With $R_2(3) = 6$ the bound gives $S(3) \le 13$, which is exact. Under the hypothesis $R_4(3) \le 61$, a computer-assisted claim of M. Tatarevic, the pigeonhole step gives $R_5(3) \le 302$, and the bound then gives $S(6) \le 1801$; the best published upper bound is $S(6) \le 1836$. We then study a Schur colouring of the longest interval that the bound allows. Let $R_k(3) \le u+1$, $(k+1)u = 2t$ and $m = (k+2)t$; then the pigeonhole step and the bound give $S(k+2) \le 2m+1$. Let c be a Schur colouring of $[1, 2m+1]$ with $k+2$ colours. Then each colour occurs exactly t times in $[1, m]$. With $q = c(m+1)$, the neighbourhood of the centre m in the colour q is regular in every other colour, and $c(m+1-d) = c(m+1+d)$ for every $d \in [1, m]$ with $c(d) = q$. For six colours under $R_4(3) \le 61$, a Schur colouring of $[1, 1801]$ gives five sets of 60 integers. Each set is closed under $x \mapsto 1800 - x$, this map has no fixed point on it, and the differences of the set have at most four colours. We do not exclude these sets, so we do not prove $S(6) \le 1800$. The theorems are formalized in Lean 4 with Mathlib, with $R_4(3) \le 61$ as an explicit hypothesis.

1. Introduction

A set S of positive integers is *sumfree* when there are no $x, y \in S$ with $x + y \in S$; the case $x = y$ is included. The Schur number $S(n)$ is the largest N such that $\{1, \dots, N\}$ is the union of n sumfree sets. Schur proved that $S(n)$ is finite [Sch16]. Only $S(1), \dots, S(5) = 1, 4, 13, 44, 160$ are known. The first three can be found by hand [Heu18]; $S(4) = 44$ is due to Baumert [Bau61] and $S(5) = 160$ to Heule [Heu18]. For $S(6)$ the published bounds are $536 \le S(6) \le 1836$. The lower bound is due to Fredricksen and Sweet [FS00]; Eliahou and Revuelta give the upper bound as the best known one [ER21].

Let $R_k(3)$ be the least N such that every colouring of the edges of the complete graph K_N with k colours has a monochromatic triangle. The inequality $S(k) \le R_k(3) - 2$ is classical [Heu18, Eli20]. The known values and bounds are $R_1(3) = 3$, $R_2(3) = 6$, $R_3(3) = 17$ [GG55], $51 \le R_4(3) \le 62$ [Chu73, FKR04], $162 \le R_5(3) \le 307$ [Exo94, Rad26] and $R_6(3) \le 1838$ [Rad26]. The upper bounds for $R_5(3)$ and $R_6(3)$ follow from $R_4(3) \le 62$ by the pigeonhole recursion $R_{k+1}(3) \le (k+1)(R_k(3) - 1) + 2$ [GG55, Rad26], and $1836 = 1838 - 2$.

M. Tatarevic has released a computer-assisted proof of $R_4(3) \le 61$ in a public repository [Tat26]. It combines a mathematical reduction, finite enumeration and 56,830 SAT formulas, and a Lean development in the repository derives the bound from the unsatisfiability of these formulas [Tat26, Chk26]. For these formulas the repository publishes solver UNSAT records, not LRAT certificates. The author's project made and checked LRAT certificates for all 56,830 formulas, with AI assistance; the record is [Chk26], an issue that the author posted in the repository on 2 October 2026. For each formula an UNSAT proof was made with CaDiCaL and converted to LRAT with drat-trim where necessary, and the checker lrat-check accepted all 56,830 LRAT proofs. A second check by Claude (Anthropic), which used none of the code of the first pipeline, verified the same LRAT proofs with the CakeML-verified checker cake_lpr (56,830 of 56,830). These checks rely on the builds of the checkers and on Tatarevic's Lean emitters that produce the formulas, and they do

not review the reduction by hand. We know of no preprint or refereed publication of this result (Section 5). In this paper $R_4(3) \le 61$ is a hypothesis, and every statement that uses it says so.

We prove the following.

Theorem 1.1. Let $k, r \ge 1$. If $R_k(3) \le r$, then

$$S(k+1) \le 2(k+1)\left\lfloor \frac{r-1}{2} \right\rfloor + 1.$$

Corollary 1.2. (a) $S(3) \le 13$. (b) If $R_4(3) \le 61$, then $R_5(3) \le 302$ and $S(6) \le 1801$.

The proof of Theorem 1.1 (Section 3) places a large part of one colour class symmetrically about the centre of the interval and applies the bound on $R_k(3)$ to the points obtained (see Section 5 on its relation to [Wan97]). From the same hypothesis $R_4(3) \le 61$, other methods give $S(6) \le 1805$ and $S(6) \le 1806$ (Section 3.3).

Let $R_k(3) \le 2t+2$ and $m = (k+1)t$. Then Theorem 1.1 gives $S(k+1) \le 2m+1$. Section 4 describes a Schur colouring of the interval $[1, 2m+1]$, which we call the *frontier interval*. Each colour occurs exactly t times in $[1, m]$ (Theorem 4.1). With $k+2$ colours and two consecutive Ramsey bounds that meet (Section 4.2), the neighbourhood of the centre in one colour is regular in each other colour (Theorem 4.4), the colouring is reflected about $m+1$ at the points of that colour (Theorem 4.7), and the neighbourhoods of the endpoint split into pairs (Theorem 4.9). For six colours this gives the following.

Theorem 1.3 (Theorem 4.11, informal). Suppose $R_4(3) \le 61$ and $S(6) \ge 1801$. Then a Schur colouring of $[1, 1801]$ with six colours gives five sets of 60 integers. Each set is closed under $x \mapsto 1800 - x$, this map has no fixed point on it, and the differences of two distinct elements of the set have at most four colours.

Theorem 1.3 does not prove $S(6) \le 1800$. Whether five sets with these properties can exist is open.

All theorems of Sections 2 to 4 are formalized in Lean 4 with Mathlib (Section 6). Section 5 lists the literature that we checked. We make no claim of novelty.

2. Definitions and lemmas

All numbers are integers, $[a, b] = \{x : a \le x \le b\}$, and $|x - y|$ is the distance of x and y.

A set X of positive integers is *covered by n sumfree sets* when there are sumfree sets $C_1, \dots, C_n$ with $X \subseteq C_1 \cup \cdots \cup C_n$. The sets need not be disjoint or contained in X. A subset of a sumfree set is sumfree, so $S(n)$ is also the largest N such that $[1, N]$ is covered by n sumfree sets. If $[1, M+1]$ is not covered by n sumfree sets, then $S(n) \le M$.

We write $R_k(3) \le r$ for the statement: for every set V of at least r points and every colouring of the pairs of points of V with at most k colours, there is a monochromatic triangle.

Let $n \ge 1$. A *Schur colouring of $[1, N]$ with n colours* is a map c from $[1, N]$ to a set of n colours such that there are no $x, y \ge 1$ with $x + y \le N$ and $c(x) = c(y) = c(x+y)$; the case $x = y$ is included.

Lemma 2.1. Let $n \ge 1$.

(a) If c is a Schur colouring of $[1, N]$ with n colours, then its n colour classes are sumfree sets that cover $[1, N]$.

(b) If the sumfree sets $C_1, \dots, C_n$ cover $[1, N]$, then a map c that gives each $x \in [1, N]$ an index i with $x \in C_i$ is a Schur colouring of $[1, N]$ with n colours.

Proof. (a) A colour class is a subset of $[1, N]$, and three elements x, y, $x + y$ of one class are excluded by the definition. (b) If $c(x) = c(y) = c(x+y) = i$, then $x, y, x+y \in C_i$, which is false. □

The difference colouring. Let c be a Schur colouring of $[1, N]$. Give each pair $\{x, y\}$ of distinct points of $[0, N]$ the colour $c(|x - y|)$; this is defined, since $1 \le |x - y| \le N$.

Lemma 2.2. The difference colouring has no monochromatic triangle.

Proof. Let $x < y < z$ be points of $[0, N]$. The numbers $y - x$, $z - y$ and $z - x$ lie in $[1, N]$, and $(y - x) + (z - y) = z - x$. So they do not all have the same colour. □

Lemma 2.3. Let c be a Schur colouring of $[1, N]$, let $V \subseteq [0, N]$, and let K be a set of at most k colours with $c(|x - y|) \in K$ for all distinct $x, y \in V$. If $R_k(3) \leq r$, then $|V| < r$.

Proof. Otherwise the difference colouring on V uses at most k colours on at least r points, and it has a monochromatic triangle. This contradicts Lemma 2.2. □

For a finite set $V \subseteq [0, N]$, a point v and a colour i, the *colour-i neighbourhood* of v in V is

$$\Gamma_i(v; V) = \{w \in V : \ w \neq v, \ c(|v - w|) = i\}.$$

The sets $\Gamma_i(v; V)$, for all colours i, partition $V \setminus \{v\}$.

Lemma 2.4. Let c be a Schur colouring of $[1, N]$, let $V \subseteq [0, N]$ and $v \in [0, N]$, and let i be a colour. If x and y are distinct points of $\Gamma_i(v; V)$, then $c(|x - y|) \neq i$.

Proof. The points v, x and y are distinct points of $[0, N]$, and the pairs $\{v, x\}$ and $\{v, y\}$ have the colour i. If $\{x, y\}$ also had the colour i, the three points would form a monochromatic triangle, which contradicts Lemma 2.2. □

Lemma 2.5 (pigeonhole step [GG55]). Let $k, N \geq 1$. If $R_k(3) \leq N$, then $R_{k+1}(3) \leq (k + 1)(N - 1) + 2$.

Proof. Let V be a set of at least $(k + 1)(N - 1) + 2$ points whose pairs have at most $k + 1$ colours, and fix $v \in V$. The other points, at least $(k + 1)(N - 1) + 1$ of them, are joined to v in at most $k + 1$ colours, so one colour i occurs on at least N of these pairs. Let U be the set of their other endpoints. If two points of U are joined in the colour i, they form a triangle of colour i with v. Otherwise the pairs of U have at most k colours, and $R_k(3) \leq N$ gives a monochromatic triangle in U. □

3. The centred bound

Theorem 3.1. Let $k, r \geq 1$ and suppose that $R_k(3) \leq r$. Put $q = \lfloor (r - 1)/2 \rfloor$ and $h = (k + 1)q + 1$. Then $[1, 2h]$ is not covered by $k + 1$ sumfree sets. So

$$S(k + 1) \leq 2h - 1 = 2(k + 1)q + 1.$$

Proof. Suppose that the sumfree sets $C_1, \ldots, C_{k+1}$ cover $[1, 2h]$. For each $x \in [1, 2h]$ choose an index $f(x)$ with $x \in C_{f(x)}$.

1. The interval $[1, h]$ has $h = (k + 1)q + 1$ elements and f takes at most $k + 1$ values on it. So there is an index i with $f(a) = i$ for at least $q + 1$ elements $a \in [1, h]$. Let A be the set of these elements. Then $A \subseteq C_i \cap [1, h]$ and $|A| \geq q + 1$.
2. Let $P = \{h - a : \ a \in A\} \cup \{h + a : \ a \in A\}$. The points $h - a$ lie in $[0, h - 1]$ and the points $h + a$ lie in $[h + 1, 2h]$. So $|P| = 2|A| \geq 2q + 2 \geq r$, because $2q \geq r - 2$.
3. Give each pair $x < y$ of points of P the colour $f(y - x)$. This is defined, because $1 \leq y - x \leq 2h$.
4. No pair has the colour i. Let $x < y$ be points of P with $y - x \in C_i$. If $x = h - a$ and $y = h + b$ with $a, b \in A$, then $y - x = a + b$, and a, b, $a + b$ lie in C_i. If $x = h - a$ and $y = h - b$ with $a, b \in A$ and $b < a$, then $y - x = a - b$ and $(a - b) + b = a$, with $a - b$, b, a in C_i. If $x = h + b$ and $y = h + a$ with $b < a$, the same holds. Each case contradicts the fact that C_i is sumfree.
5. So the pairs of P have at most k colours. Since $|P| \geq r$ and $R_k(3) \leq r$, there are $x < y < z$ in P and an index j with $f(y - x) = f(z - y) = f(z - x) = j$. Then $y - x$, $z - y$ and $(y - x) + (z - y) = z - x$ lie in C_j, which contradicts the fact that C_j is sumfree.

So $[1, 2h]$ is not covered by $k + 1$ sumfree sets, and $S(k + 1) \leq 2h - 1$. □

3.1 Two cases

Corollary 3.2. $S(3) \le 13$; that is, $[1, 14]$ is not covered by three sumfree sets. Since $S(3) = 13$, the bound is exact.

Proof. Take $k = 2$ and $r = 6$ in Theorem 3.1; $R_2(3) \le 6$ follows from $R_1(3) \le 3$ by Lemma 2.5. Then $q = 2$, $h = 7$ and $2h = 14$. □

Corollary 3.3. Suppose that $R_4(3) \le 61$. Then $R_5(3) \le 302$ and $S(6) \le 1801$; that is, $[1, 1802]$ is not covered by six sumfree sets.

Proof. Lemma 2.5 with $k = 4$ and $N = 61$ gives $R_5(3) \le 5 \cdot 60 + 2 = 302$. Theorem 3.1 with $k = 5$ and $r = 302$ gives $q = 150$, $h = 901$ and $2h = 1802$. □

3.2 Values of the bound

For even r the bound of Theorem 3.1 is $(k+1)(r-2)+1$, and for odd r it is $(k+1)(r-1)+1$. The route through $S(k+1) \le R_{k+1}(3) - 2$ and Lemma 2.5 gives $S(k+1) \le (k+1)(r-1)$ from $R_k(3) \le r$. So Theorem 3.1 gives a bound that is smaller by k when r is even and larger by 1 when r is odd.

Table 1 lists the bound of Theorem 3.1 for known Ramsey bounds. Every number in the third column comes from Theorem 3.1. The last column says which rows are stated as Lean theorems (Section 6).

Table 1. The bound of Theorem 3.1.

colours	Ramsey bound used	Theorem 3.1 gives	known value	Lean
2	$R_1(3) = 3$	$S(2) \le 5$	$S(2) = 4$	instance not stated
3	$R_2(3) = 6$	$S(3) \le 13$	$S(3) = 13$	Corollary 3.2
4	$R_3(3) = 17$ [GG55]	$S(4) \le 65$	$S(4) = 44$ [Bau61]	instance not stated
5	$R_4(3) \le 62$ [FKR04]	$S(5) \le 301$	$S(5) = 160$ [Heu18]	Ramsey bound not formalized
6	$R_5(3) \le 307$ [Rad26]	$S(6) \le 1837$	open	Ramsey bound not formalized
6	$R_5(3) \le 302$, from $R_4(3) \le 61$	$S(6) \le 1801$	open	Corollary 3.3

3.3 Comparison for $S(6)$

The following list gives upper bounds for $S(6)$ and labels each by its origin. Only the first item and the item for 1837 come from this paper.

- $S(6) \le 1801$ under $R_4(3) \le 61$. This paper, Corollary 3.3, Lean theorem `not_coveredBySumFree_Icc_six_of_triangleRamsey_four_sixtyOne`.
- $S(6) \le 1805$ under $R_4(3) \le 61$. Another method, for comparison: the pigeonhole recursion with the parity refinement [GG55; Rad26, Section 6.1]. From $R_5(3) \le 302$, suppose that the edges of K_{1807} have a colouring with six colours and no monochromatic triangle. A vertex has at most 301 neighbours in each colour: among 302 such neighbours there is an edge of that colour, or their edges have at most five colours, and both cases give a monochromatic triangle. Since each vertex has $1806 = 6 \cdot 301$ neighbours, it has exactly 301 in each colour. Then each colour class is a 301-regular graph on 1807 vertices, which is impossible because $1807 \cdot 301$ is odd. So $R_6(3) \le 1807$ and $S(6) \le 1805$. This is not formalized.
- $S(6) \le 1806$ under $R_4(3) \le 61$. Another method, for comparison: Eliahou's adaptive bound [Eli20, Proposition 4], which is [Eli19, Proposition 2.7] in the preprint, states $R_n(3) \le n!(e - a/24) + 1$ for $n \ge 4$ when $a \le 66 - R_4(3)$. With $a = 5$ and $n = 6$ it gives $R_6(3) \le 1808$, so $S(6) \le 1806$. The pigeonhole recursion without the parity refinement gives the same values. This is not formalized.
- $S(6) \le 1836$. Published [ER21], from $R_6(3) \le 1838$ [Rad26]; another method, for comparison. It uses only the published bound $R_4(3) \le 62$.

- $S(6) \le 1837$ from the published bound $R_5(3) \le 307$. This paper, Theorem 3.1 with an odd r; it is weaker than 1836.

So under $R_4(3) \le 61$ the bound of Corollary 3.3 is 4 below the parity route and 5 below the bound from [Eli20]. With the published bound $R_4(3) \le 62$ it gives nothing new for $S(6)$.

4. The frontier

Let $k, t \ge 1$, suppose that $R_k(3) \le 2t+2$, and let $m = (k+1)t$. Theorem 3.1 with $r = 2t+2$ gives $q = t$ and $S(k+1) \le 2m+1$. This section describes a Schur colouring of the frontier interval $[1, 2m+1]$. The results are necessary conditions; they do not decide whether such a colouring exists. For example, $R_2(3) = 6$ gives $t = 2$, $m = 6$ and the frontier interval $[1, 13]$ for three colours, which has Schur colourings because $S(3) = 13$.

4.1 Balanced colour classes

Theorem 4.1. Let $R_k(3) \le 2t+2$, let $m = (k+1)t$, and let c be a Schur colouring of $[1, 2m+1]$ with $k+1$ colours. Then each colour occurs exactly t times in $[1, m]$.

Proof. For a colour j, let $D_j = \{d \in [1, m] : c(d) = j\}$. The points $m - d$ and $m + d$ with $d \in D_j$ are $2|D_j|$ distinct points of $[0, 2m]$, and they lie in $\Gamma_j(m; [0, 2m+1])$, because their distance to m is d. By Lemma 2.4 the differences of these points avoid the colour j, so they have at most k colours. Lemma 2.3 gives $2|D_j| < 2t+2$, so $|D_j| \le t$. The $k+1$ sets D_j partition $[1, m]$, which has $(k+1)t$ elements. So $|D_j| = t$ for every j. □

Remark 4.2 (two centres). The reflection $x \mapsto N - x$ of $[0, N]$ keeps all distances. So, for any colouring c, the points v and $N - v$ have the same number of neighbours of each colour in $[0, N]$. For $N = 2m+1$ this applies to the two centres m and $m+1$. This remark is proved in Lean but is not among the compared statements (Section 6).

4.2 The central neighbourhood and nested saturation

From here on we use $k+2$ colours and the following hypotheses.

Frontier setting. Let $k, u, t \ge 1$ and m satisfy

$$R_k(3) \le u + 1, \qquad 2t = (k+1)u, \qquad m = (k+2)t,$$

and put $N = 2m+1$. Let c be a Schur colouring of $[1, N]$ with $k+2$ colours, let $q = c(m+1)$, and let

$$V = \Gamma_q(m; [0, N])$$

be the *central neighbourhood*. The endpoint N lies in V, because $N - m = m + 1$. The point 0 lies in V when $c(m) = q$.

By Lemma 2.5, $R_{k+1}(3) \le (k+1)u + 2 = 2t + 2$. So the Ramsey bound for k colours, through Lemma 2.5, meets the bound for $k+1$ colours that Theorem 4.1 needs, and $[1, N]$ is the frontier interval for $k+2$ colours. For three colours, the values $k = 1$, $u = 2$, $t = 2$ and $m = 6$ satisfy these hypotheses, with $N = 13$.

Theorem 4.3. In the frontier setting, $|V| = 2t + 1$.

Proof. Theorem 4.1, with $k+1$ in place of k, applies to c and gives t points $d \in [1, m]$ with $c(d) = q$. The $2t$ points $m \pm d$ for these d lie in $V \cap [0, 2m]$, and $N \in V$. So $|V| \ge 2t + 1$. By Lemma 2.4 at m, the differences of V avoid the colour q, so they have at most $k+1$ colours. Lemma 2.3 with $R_{k+1}(3) \le 2t+2$ gives $|V| < 2t + 2$. □

Theorem 4.4 (nested saturation). In the frontier setting, let $v \in V$. Then $\Gamma_q(v; V)$ is empty, and $|\Gamma_i(v; V)| = u$ for every colour $i \ne q$.

Proof. By Lemma 2.4 at m, no two points of V have a difference of colour q, so $\Gamma_q(v; V)$ is empty. Let $i \ne q$. The differences of $\Gamma_i(v; V)$ avoid the colour i (Lemma 2.4 at v) and the colour q (Lemma 2.4 at m).

So they have at most k colours, and Lemma 2.3 with $R_k(3) \le u+1$ gives $|\Gamma_i(v;V)| \le u$. The sets $\Gamma_i(v;V)$ partition $V \setminus \{v\}$, which has $2t = (k+1)u$ points by Theorem 4.3. There are $k+1$ colours $i \ne q$, and each of their neighbourhoods has at most u points. So each has exactly u points. □

4.3 Automorphism extension

The next lemma is about an arbitrary colouring of pairs.

Lemma 4.5. Let col be a map that gives each ordered pair of elements of a set a colour. Let W be a finite subset, e an element not in W, and J a map with $J(W) \subseteq W$, $J(J(x)) = x$ and $\mathrm{col}(J(x), J(y)) = \mathrm{col}(x,y)$ for all $x, y \in W$. Let $v \in W$, and suppose that for each colour γ the number of $w \in (W \cup \{e\}) \setminus \{v\}$ with $\mathrm{col}(v,w) = \gamma$ is equal to the number of $w \in (W \cup \{e\}) \setminus \{J(v)\}$ with $\mathrm{col}(J(v),w) = \gamma$. Then $\mathrm{col}(v,e) = \mathrm{col}(J(v),e)$.

Proof. For each colour γ, the map J is a bijection from $\{w \in W \setminus \{v\} : \mathrm{col}(v,w) = \gamma\}$ onto $\{w \in W \setminus \{J(v)\} : \mathrm{col}(J(v),w) = \gamma\}$, with inverse J. So the two counts inside W are equal for each colour. Take $\gamma = \mathrm{col}(v,e)$. The count for v in $W \cup \{e\}$ is the count in W plus 1. If $\mathrm{col}(J(v),e) \ne \gamma$, the count for $J(v)$ in $W \cup \{e\}$ is the count in W, which contradicts the hypothesis. □

4.4 Forced reflection

Lemma 4.6. Let c be any map, $m \ge 0$, and $V = \Gamma_{c(m+1)}(m; [0, 2m+1])$. If $x \in V$ and $x \ne 2m+1$, then $2m - x \in V$ and $2m - x \ne 2m+1$.

Proof. Since $x \le 2m$, the point $2m - x$ lies in $[0, 2m]$. It has the same distance to m as x, so $2m - x \ne m$ and $c(|m - (2m-x)|) = c(m+1)$. □

Theorem 4.7 (forced reflection). In the frontier setting, let $d \in [1, m]$ with $c(d) = q$. Then $c(m+1-d) = c(m+1+d)$.

Proof. Let $W = V \setminus \{N\}$, $e = N$, $J(x) = 2m - x$, and $\mathrm{col}(x,y) = c(|x-y|)$ for $x \ne y$; the pairs (x,x) get one fixed colour and play no part. By Lemma 4.6, J maps W to W. It is an involution, and $|J(x) - J(y)| = |x-y|$, so it keeps the colours of all pairs in W. The point $v = m + d$ lies in W, since $v \ne m$, $c(|v-m|) = c(d) = q$ and $v \le 2m$; and $J(v) = m - d$. Here $W \cup \{e\} = V$. By Theorem 4.4, both v and $J(v)$ have u neighbours in V of each colour $i \ne q$ and none of colour q. So Lemma 4.5 applies and gives $c(|m+d-N|) = c(|m-d-N|)$, that is, $c(m+1-d) = c(m+1+d)$. □

Remark 4.8. The reflection is not a property of every Schur colouring. The colouring of $[1,5]$ with the colour classes $\{1,4\}$, $\{2,3\}$ and $\{5\}$ is a Schur colouring with three colours. Take $m = 2$, so that $[1,5] = [1, 2m+1]$. Then $q = c(3)$ is the colour of $\{2,3\}$, and $d = 2$ has $c(d) = q$, but $c(m+1-d) = c(1) \ne c(5) = c(m+1+d)$. In Theorem 4.7 the reflection comes from the hypotheses of the frontier setting.

4.5 Paired endpoint neighbourhoods and parity

In the frontier setting, for a colour $i \ne q$ let

$$P_i = \Gamma_i(N; V)$$

be the *endpoint neighbourhood* of colour i.

Theorem 4.9. In the frontier setting, let $i \ne q$. Then:

(a) $|P_i| = u$;

(b) for every $x \in P_i$, the point $2m - x$ lies in P_i and $2m - x \ne x$;

(c) for distinct $x, y \in P_i$, the colour $c(|x-y|)$ is neither i nor q.

Proof. (a) This is Theorem 4.4 at $v = N$. (b) Let $x \in P_i$. Then $x \in V$ and $x \ne N$, so $x \ne m$, $x \le 2m$ and $d = |m-x|$ satisfies $1 \le d \le m$ and $c(d) = q$. If $x = m-d$, then $N - x = m+1+d$ and $N - (2m-x) = m+1-d$.

If $x = m + d$, then $N - x = m + 1 - d$ and $N - (2m - x) = m + 1 + d$. In both cases Theorem 4.7 gives $c(N - (2m - x)) = c(N - x) = i$. By Lemma 4.6, $2m - x \in V \setminus \{N\}$, so $2m - x \in P_i$. Since $x \neq m$, we have $2m - x \neq x$. (c) The colour is not i by Lemma 2.4 at N, and it is not q by Lemma 2.4 at m, because $P_i \subseteq V$. □

Corollary 4.10. In the frontier setting, u is even. So if u is odd, then no Schur colouring of $[1, 2m + 1]$ with $k + 2$ colours exists.

Proof. There is a colour $i \neq q$, since $k + 2 \geq 2$. By Theorem 4.9, P_i lies in $[0, 2m] \setminus \{m\}$, and the map $x \mapsto 2m - x$ is a bijection from $P_i \cap [0, m - 1]$ onto $P_i \cap [m + 1, 2m]$. So $u = |P_i| = 2\,|P_i \cap [0, m - 1]|$. □

4.6 Six colours under $R_4(3) \leq 61$

Theorem 4.11. Suppose that $R_4(3) \leq 61$. Let c be a Schur colouring of $[1, 1801]$ with six colours, and let $q = c(901)$. Let V be the set of points $x \in [0, 1801]$ with $x \neq 900$ and $c(|900 - x|) = q$, and for a colour $i \neq q$ let P_i be the set of points $x \in V$ with $x \neq 1801$ and $c(1801 - x) = i$. Then:

(a) each colour occurs exactly 150 times in $[1, 900]$;

(b) $|V| = 301$, and each $v \in V$ has exactly 60 neighbours $w \in V$, $w \neq v$, with $c(|v - w|) = i$, for each colour $i \neq q$;

(c) $c(901 - d) = c(901 + d)$ for every $d \in [1, 900]$ with $c(d) = q$;

(d) for each colour $i \neq q$, the set P_i has 60 points; it is closed under $x \mapsto 1800 - x$, and this map has no fixed point on P_i; and for distinct $x, y \in P_i$ the colour $c(|x - y|)$ is neither i nor q.

Proof. The frontier setting holds with $k = 4$, $u = 60$, $t = 150$ and $m = 900$: $R_4(3) \leq 61 = u + 1$, $2t = 300 = 5 \cdot 60$, $m = 6 \cdot 150$ and $N = 1801$. Part (a) is Theorem 4.1 with $R_5(3) \leq 302$ (Corollary 3.3). Part (b) is Theorems 4.3 and 4.4, part (c) is Theorem 4.7, and part (d) is Theorem 4.9. □

By part (d), each P_i is the union of 30 pairs $\{900 - d, 900 + d\}$ with $c(d) = q$. The difference colouring on P_i is a colouring of the edges of a complete graph on 60 vertices with at most four colours and no monochromatic triangle (Lemma 2.2), and the map $x \mapsto 1800 - x$ is an involution of P_i without fixed point that keeps the colours of all edges. So, under $R_4(3) \leq 61$, a Schur colouring of $[1, 1801]$ with six colours gives five such structures, one for each colour $i \neq q$.

Remark 4.12. Let c be a Schur colouring of $[1, 1801]$ with six colours, and suppose that $R_4(3) \leq 61$. Then $R_4(3) = 61$. Indeed, let $q = c(901)$ and take a colour $i \neq q$. The 60 points of P_i lie in $[0, 1800]$, and by Theorem 4.11 (d) and Lemma 2.2 the difference colouring on P_i is a colouring of the edges of K_{60} with at most four colours and no monochromatic triangle. So $R_4(3) \geq 61$. This remark is not formalized.

These results do not prove $S(6) \leq 1800$, even under $R_4(3) \leq 61$. They give necessary conditions for the existence of a Schur colouring of $[1, 1801]$ with six colours, and we do not show that they cannot be met. Since $u = 60$ is even, Corollary 4.10 does not apply. Whether the five 60-point structures of Theorem 4.11 exist is open.

5. Literature

We checked the following sources, first in a local library of papers and then on the web, in September and October 2026. We looked for Theorem 3.1, the argument of its proof, an upper bound for $S(6)$ below 1836, and a consequence of $R_4(3) \leq 61$ for the Schur numbers.

- Eliahou and Revuelta, both the preprint [ER20] and the refereed version [ER21]. Both give $S(6) \leq R_6(3) - 2 \leq 1836$ as the best known upper bound, and both say that, as far as theory is concerned, nothing better than $S(4) \leq R_4(3) - 2 \leq 60$ is known.
- Eliahou, both the preprint [Eli19] and the refereed version [Eli20]. The bound 1806 of Section 3.3 comes from [Eli20, Proposition 4], which is [Eli19, Proposition 2.7].

- Heule [Heu18]; Irving [Irv73]; Radziszowski [Rad26]; Hegde, Lott, Petridis and Ponagandla [HLPP26]; Abbott and Hanson [AH72], in a scan with poor text recognition.
- Adhikari, Boza, Eliahou, Marín, Revuelta and Sanz [ABEMRS15]. It cites the Schur bound of [Wan97] in one sentence (Section 1.1), with no method. In the proof of its Corollary 5.7 it uses $S_2(5) \leq 306$, that is, $S(5) \leq 305$.
- Eliahou and Fares [EF16]. It cites [Wan97] only for the bound on $R_n(3)$.
- Myers [Mye15], a PhD thesis. It cites [Wan97] once, only for the constant of the bound on $R_n(3)$, and it does not mention the Schur bound of [Wan97].
- Kościuszko [Kos25]. For another equation, it splits a colour neighbourhood of a vertex v into the parts $v + A$ and $v - A$. It uses the larger part only in a lower bound for the independence number, and it takes a largest independent set as the next set. It uses no centre point and no parity step.
- The repository [Tat26], all its Markdown files at the commit cited. They do not mention Schur numbers. We found no preprint or other announcement of $R_4(3) \leq 61$ on arXiv, Hacker News, MathOverflow, Mathematics Stack Exchange, Tatarevic's blog or in the OEIS entry A003323. We could not search X, Reddit, Bluesky or Mastodon.

None of the sources that we read states Theorem 3.1, its argument, an upper bound for $S(6)$ below 1836, or a consequence of $R_4(3) \leq 61$ for the Schur numbers.

From the published even bound $R_4(3) \leq 62$, Theorem 3.1 gives $S(5) \leq 301$ (Table 1). [ER21, Section 5.2] calls $S(5) \leq 305$ "the currently best known theoretical upper bound", and [ABEMRS15] uses the same value 305. The bound of [Wan97] on S_n that is discussed below is stated only for even $n \geq 6$.

Two sources may contain the argument of Theorem 3.1 or a similar argument, and we did not read either of them in full. By its zbMATH summary (Zbl 0882.05095), Wan [Wan97] proves $R_n(3) < n!\,(e-e^{-1}+3)/2+1$ for $n \geq 4$, which [Eli20] also cites, and $S_n < n!\,(e-e^{-1}+3)/2-n+2$ for even $n \geq 6$. We could not obtain the full text of [Wan97], so we do not know its argument. The summary does not define S_n. Two conventions are in use [Heu18, footnote 1]: S_n is the largest N such that $[1, N]$ has a Schur colouring with n colours, which is our $S(n)$, or the least N such that every colouring of $[1, N]$ with n colours has a monochromatic solution of $x + y = z$, which is $S(n) + 1$. Adhikari et al. [ABEMRS15] use the second convention and write $S_2(n)$. From Wan's bound $R_5(3) \leq 322$, the route through Lemma 2.5 gives $S(6) \leq 1926$, and Theorem 3.1 gives $S(6) \leq 1921$.

- If Wan's S_n is our $S(n)$, the summary gives $S(6) \leq 1922$, which is 1 above the value 1921 that Theorem 3.1 gives from Wan's own bound $R_5(3) \leq 322$.
- If Wan's S_n is $S(n) + 1$, the summary gives $S(6) \leq 1921$, the same value. Under this reading the two bounds agree for every even n from 6 to 30: put $r = \lfloor (n-1)!\,(e - e^{-1} + 3)/2 \rfloor + 1$, which is Wan's bound on $R_{n-1}(3)$ and is even for these n; then $\lfloor n!\,(e - e^{-1} + 3)/2 \rfloor - n + 2 - 1 = 2n \lfloor (r-1)/2 \rfloor + 1$. We computed both sides in exact rational arithmetic.

So Wan's bound for even n may come from the argument of Theorem 3.1, or from an argument of the same strength. We have not read his proof. The method of Theorem 3.1 is probably not new.

By its abstract, Li Huai'en [Li92] proves $R_n(3) \leq n!\,(3/2 + \sinh 1) + 1$, where $3/2 + \sinh 1 = (e - e^{-1} + 3)/2$, with the pigeonhole principle and parity arguments, and also improves upper bounds for the Schur numbers. We read only its abstract.

We did not read Whitehead [Whi73]; Xu, Xie and Chen [XXC02], whose title names upper bounds for $R_n(3)$ and for the Schur numbers; Fredricksen [Fre79]; Landman and Robertson [LR14]; or Soifer [Soi24]. In October 2026 OpenAlex listed 24 works that cite [Wan97]; apart from the sources listed above, we did not go through them. This list of unread sources is not complete.

We make no claim of novelty for Theorem 3.1 or for its method.

6. Lean formalization

The theorems of Sections 2 to 4 are formalized in Lean 4 [MU21] with Mathlib [mC20], with the Lean toolchain v4.35.0-rc3 and the Mathlib tag v4.35.0-rc3. The library `ClassicalSchur` has four modules:

- `Basic`: sumfree sets (`SumFree`) and covers by sumfree sets (`CoveredBySumFree`);
- `Ramsey`: the statement `TriangleRamsey k N`, which is $R_k(3) \le N$; the pigeonhole bound $\rho(k)$ (`ramseyBound`) with $\rho(0) = 2$ and $\rho(k+1) = (k+1)(\rho(k)-1)+2$; and the theorem $R_k(3) \le \rho(k)$ for every natural k (`triangleRamsey_ramseyBound`);
- `SchurBound`: Lemma 2.5, Theorem 3.1 and Corollaries 3.2 and 3.3;
- `Frontier`: Schur colourings, colour neighbourhoods, Lemmas 2.1 to 2.4, and Section 4.

The file `Challenge.lean` imports only Mathlib. It states seven definitions (`SumFree`, `CoveredBySumFree`, `TriangleRamsey`, `SchurColoring`, `colorNbhd`, `centralNbhd`, `endpointNbhd`) and fourteen theorems, with the proofs left as `sorry`. The file `Solution.lean` imports the library, which proves the fourteen theorems under the same names. The file `comparator.json` lists the fourteen theorems and permits only the axioms `propext`, `Classical.choice` and `Quot.sound`. The script `scripts/verify-comparator.sh` runs `lake comparator` [LFRO], which is part of Lake in the Lean toolchain, on `Challenge.lean`, `Solution.lean` and a copy of `comparator.json`, to which it adds the kernels `nanoda_bin` and `con-ron` of the toolchain. The comparator compares each of the fourteen theorems, which `Solution.lean` imports from the library, with the definitions that it uses, against `Challenge.lean`, and checks the axioms. The repository's continuous integration runs this script. The fourteen theorems use no axioms other than `propext`, `Classical.choice` and `Quot.sound`.

The hypothesis $R_4(3) \le 61$ enters only as the explicit argument `TriangleRamsey 4 61` of two theorems:

- `not_coveredBySumFree_Icc_six_of_triangleRamsey_four_sixtyOne` (Corollary 3.3);
- `schur_six_frontier_structure` (Theorem 4.11).

The library does not prove $R_4(3) \le 61$ and does not import [Tat26]. The Lean development of [Tat26] uses its own definitions and Lean v4.32.1, and this repository does not join the two developments in one build.

The results correspond to Lean declarations in the namespace `ClassicalSchur` as follows. The declarations marked $*$ are among the fourteen compared theorems.

Table 2. Results and Lean declarations.

Result	Lean declaration
Lemma 2.1 (a)	`coveredBySumFree_of_schurColoring` $*$
Lemma 2.1 (b)	`exists_schurColoring_of_coveredBySumFree` $*$
Lemma 2.2	`SchurColoring.not_mono`
Lemma 2.3	`SchurColoring.card_lt`
Lemma 2.4	`SchurColoring.color_ne_of_mem_colorNbhd`
Lemma 2.5	`triangleRamsey_succ` $*$
Theorem 3.1	`not_coveredBySumFree_Icc_of_triangleRamsey` $*$
Corollary 3.2	`not_coveredBySumFree_Icc_fourteen_three` $*$
Corollary 3.3	`not_coveredBySumFree_Icc_six_of_triangleRamsey_four_sixtyOne` $*$
Theorem 4.1	`card_filter_Icc_eq_of_frontier` $*$
Remark 4.2	`card_colorNbhd_range_reflect`, `card_colorNbhd_two_centres`
Theorem 4.3	`card_centralNbhd_of_frontier` $*$
Theorem 4.4	`card_colorNbhd_centralNbhd_of_frontier` $*$, `SchurColoring.color_ne_of_mem_centralNbhd`
Lemma 4.5	`color_eq_of_card_filter_eq` $*$
Lemma 4.6	`reflect_mem_centralNbhd`
Theorem 4.7	`color_reflect_of_frontier` $*$
Theorem 4.9	`endpointNbhd_of_frontier` $*$
Corollary 4.10	`even_of_frontier` $*$, `not_schurColoring_frontier_of_odd`

Result	Lean declaration
Theorem 4.11	`schur_six_frontier_structure` ∗

The library has eight more public theorems, which have no number in this paper:

- `triangleRamsey_ramseyBound` ($R_k(3) \le \rho(k)$ for every natural k), `two_le_ramseyBound` ($\rho(k) \ge 2$), `ramseyBound_three` ($\rho(3) = 17$) and `ramseyBound_four` ($\rho(4) = 66$);
- `mem_colorNbhd` and `mem_centralNbhd`, which state when a point lies in $\Gamma_i(v; V)$ and in the central neighbourhood;
- `sum_card_colorNbhd`: the numbers $|\Gamma_i(v; V)|$, for all colours i, have the sum $|V \setminus \{v\}|$. Section 2 states this as a partition of $V \setminus \{v\}$;
- `endpoint_mem_centralNbhd`: the endpoint $2m + 1$ lies in the central neighbourhood (Section 4.2).

Four more theorems are private to the module `Frontier`. They are steps of the proofs of Lemma 2.2 and of Theorems 4.1, 4.3, 4.4 and 4.11.

In Corollary 3.3 the bound $R_5(3) \le 302$ is the case $k = 4$, $N = 61$ of `triangleRamsey_succ`; the Lean theorem of Corollary 3.3 states only that $[1, 1802]$ is not covered by six sumfree sets. In Corollary 3.2 the bound $R_2(3) \le 6$ is proved in Lean through $\rho(2) = 6$ and `triangleRamsey_ramseyBound`. The following are not formalized:

- the definition of $S(n)$ and the statements of Section 2 that connect it with covers by sumfree sets;
- the rows of Table 1 other than Corollaries 3.2 and 3.3;
- the closed forms and the comparison of Section 3.2, and the same comparison for even r in the abstract;
- the bounds 1805, 1806 and 1836 of Section 3.3;
- the cases $S(k + 1) \le 2m + 1$ and $S(k + 2) \le 2m + 1$ of Theorem 3.1 that define the frontier interval (abstract and Sections 1, 4 and 4.2); they are not stated as separate Lean theorems;
- the values and bounds of $S(n)$ and $R_k(3)$ that we cite from the literature, and with them the exactness of Corollary 3.2 and the examples with three colours in Sections 4 and 4.2. Without a hypothesis, Lean proves only the Ramsey bounds $R_k(3) \le \rho(k)$; it does not prove $R_4(3) \le 62$, $R_4(3) \le 61$ or $R_5(3) \le 307$;
- the statement of Section 4.2 that $0 \in V$ when $c(m) = q$;
- the paragraph after Theorem 4.11, and Remarks 4.8 and 4.12;
- the numbers of Section 5;
- the statements about vacuous cases in the list below.

The Lean statements differ from the prose in the following ways.

- The ambient type is $\mathbb{N}$. All sets are sets of natural numbers. In the statements about covers, $[1, N]$ is `Set.Icc 1 N`. In Theorems 4.1 and 4.11 (a) the interval $[1, m]$ is `Finset.Icc 1 m`, and $[0, N]$ is `Finset.range (N + 1)`. Subtraction and division are those of $\mathbb{N}$; for example, Theorem 3.1 is stated for `Set.Icc 1 (2 * ((k + 1) * ((r - 1) / 2) + 1))`, for all natural k and r. Lemma 2.5 is stated for all natural k and N, with the bound `(k + 1) * (N - 1) + 2`, and Lemma 2.1 (a) for all natural n, while the prose has $k, r, N, n \ge 1$.
- A colouring with n colours takes values in `Fin n`. In `TriangleRamsey k N` the vertices are natural numbers in a finite set of at least N elements, only the pairs $x < y$ are coloured, and the colours are natural numbers from a set of at most k elements.
- $S(n)$ is not defined in Lean. Each bound $S(n) \le M$ is stated as `¬ CoveredBySumFree (Set.Icc 1 (M + 1)) n`.
- A Schur colouring is a map from $\mathbb{N}$ to `Fin n`, defined on all of $\mathbb{N}$. Only its values on $[1, N]$ are constrained.
- The distance $|x - y|$ is `Nat.dist x y`.
- In Lemmas 2.3 and 2.4 the set $V \subseteq [0, N]$ is a `Finset` of natural numbers with the hypothesis $x \le N$ for every $x \in V$. In Lemma 2.3 the set K is a `Finset (Fin n)`.

- Neighbourhoods are `Finset`s. The central neighbourhood `centralNbhd c m` is the colour-$c(m+1)$ neighbourhood of m in $\{0, 1, \dots, 2m+1\}$, and `endpointNbhd c m i` is the colour-i neighbourhood of $2m+1$ in it.
- Lemma 2.1 says that the colour classes of a Schur colouring are sumfree sets that cover $[1, N]$, and that a map that picks an index of a covering set is a Schur colouring. The Lean theorems `coveredBySumFree_of_schurColoring` and `exists_schurColoring_of_coveredBySumFree` state only that a cover by n sumfree sets, or a Schur colouring with n colours, exists.
- Theorem 4.4 has two parts. The compared theorem `card_colorNbhd_centralNbhd_of_frontier` states only $|\Gamma_i(v; V)| = u$ for $i \neq q$. That $\Gamma_q(v; V)$ is empty follows from `SchurColoring.color_ne_of_mem_centralNbh` (no two distinct points of V have a difference of colour q), which the library proves but which is not among the compared theorems.
- `color_eq_of_card_filter_eq` (Lemma 4.5) is stated for any types with decidable equality and a colouring of ordered pairs.
- Theorem 4.1 takes the hypothesis $m = (k+1)t$ as an equation of natural numbers, with colours in `Fin (k + 1)`. It holds in Lean for all natural numbers k and t, while Section 4 has $k, t \geq 1$. The added cases are vacuous or trivial: for $t = 0$ and $k \geq 1$ the hypothesis `TriangleRamsey k 2` is false (take two points); for $k = 0$ and $t \geq 1$ the interval $[1, 2t+1]$ has no Schur colouring with one colour, since $1 + 1 = 2$; and for $k = t = 0$ the interval $[1, m]$ is empty.
- The theorems of the frontier setting (Theorems 4.3, 4.4, 4.7 and 4.9 and Corollary 4.10) take the hypotheses $2t = (k+1)u$ and $m = (k+2)t$ as equations of natural numbers, with colours in `Fin (k + 2)`. They hold in Lean for all natural numbers k, u and t, while the prose has $k, u, t \geq 1$. The added cases are vacuous: for $u = 0$ the hypothesis `TriangleRamsey k 1` is false (take a single point); for $k = 0$ and $u \geq 1$ the hypotheses give $t \geq 1$ and $m = 2t$, and $[1, 2m+1]$ has no Schur colouring with two colours, since $S(2) = 4$; and for $t = 0$ with $k, u \geq 1$ the equation $2t = (k+1)u$ is false.

Data and code availability. The Lean source, `Challenge.lean`, `Solution.lean` and `comparator.json` are in the repository that contains this paper.

Acknowledgements and use of AI tools

The argument of Theorem 3.1 was first written by an AI agent based on ChatGPT (OpenAI) in a project discussion on 27 September 2026. A Claude agent (Anthropic) audited it: it rebuilt each step, and it checked $S(2) = 4$ and the absence of a Schur colouring of $[1, 14]$ with three colours by exhaustive search. Claude (model Claude Opus 5.5) wrote the Lean proof, under the direction of the author. The argument of Section 4 (balanced colour classes, saturation, reflection, and the paired endpoint neighbourhoods with the parity of u) was proposed by an AI agent based on ChatGPT (OpenAI) in a project discussion on 2 October 2026. Claude checked each step, restated it with explicit hypotheses, and wrote the Lean proofs. The Lean kernel checks every proof. On 2 October 2026 an independent Claude agent rebuilt the Lean module of Section 4, ran its axiom audit again, and checked its statements and each step against the argument, including the example with $S(3) = 13$. On 3 October 2026 another independent Claude agent checked all fourteen compared statements and all proofs of Sections 2 to 4 of this paper against the Lean, and recomputed the numbers. The comparison of `Challenge.lean` with `Solution.lean` (the types and values of the seven definitions and the types of the fourteen theorems) was done by the Claude session that wrote the repository, not by a separate agent; the comparator run in continuous integration is the formal check of it. On 11 October 2026 a Claude agent (model Claude Opus 5.5) compared this paper with the Lean library again and completed the lists of Section 6. An independent Claude agent then checked the paper against the library, and the five points that it found were corrected. No numbered statement changed. The text of this paper was written with Claude under the direction of the author. The author has checked the content and takes responsibility for it.

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